

Tuesday, February 3

- Reading available for every lecture
- Typos/mistakes now have to be submitted after class
- Research talk Friday 11-12:15 in Davidson Lecture Hall

Roadmap

ML used to solve problems like:

- predict temperature from present weather,
- classify objects in an image,
- generate the next word in a sentence
- make an image from text description

} supervised learning
(= "right" answer)

} unsupervised learning
(no right answer)

Supervised Learning

Input: Training data

$$\begin{array}{l} x^{(1)} \in \mathbb{R}^d \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{array} \quad \begin{array}{l} y^{(1)} \in \mathbb{R} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{array}$$

← e.g., weather,
image classification,
stock prices

Output: $f: \mathbb{R}^d \rightarrow \mathbb{R}$ s.t.

$$f(x^{(i)}) \approx y^{(i)}$$

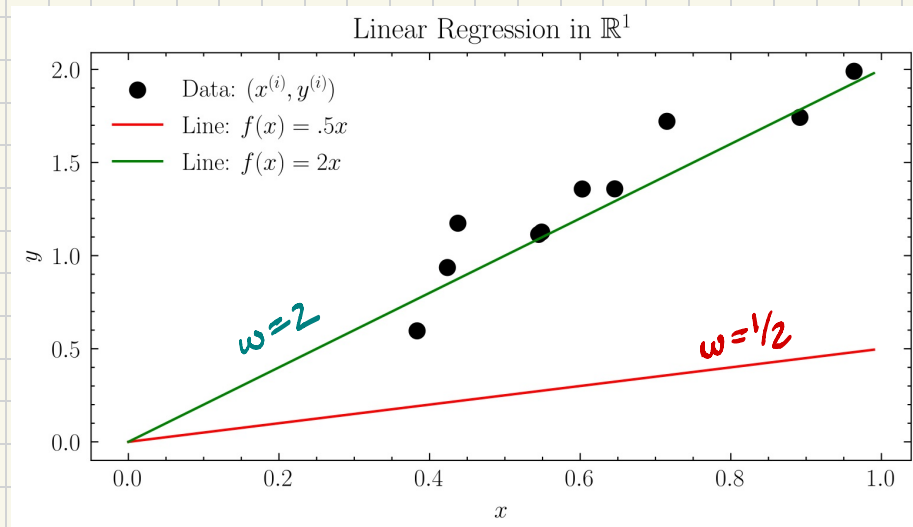
Our Strategy

- ① Function class / model
- ② Loss
- ③ Optimizer

Univariate Linear Regression

$x^{(1)}, \dots, x^{(n)} \in \mathbb{R}$ i.e., single variables

① Model: $f(x) = wx$ for some $w \in \mathbb{R}$



Q: Which line is better?

Let's formalize...

Attempt #1: $y^{(i)} - f(x^{(i)})$

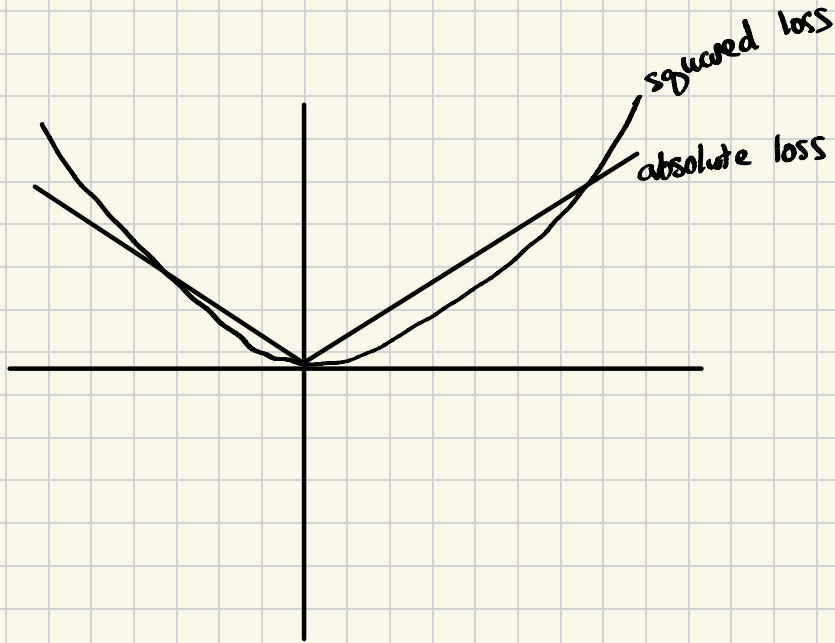
Attempt #2: $|y^{(i)} - f(x^{(i)})|$
↑ absolute loss

Attempt #3: $(y^{(i)} - f(x^{(i)}))^2$
↑ squared loss

② Loss: Mean Squared Error (MSE) Loss

$$\mathcal{L}(w) = \frac{1}{n} \sum_{i=1}^n (y^{(i)} - f(x^{(i)}))^2$$

- differentiable
- penalizes bad predictions more



③ Optimization: Derive Optimal w

Q: How do we optimize convex function?

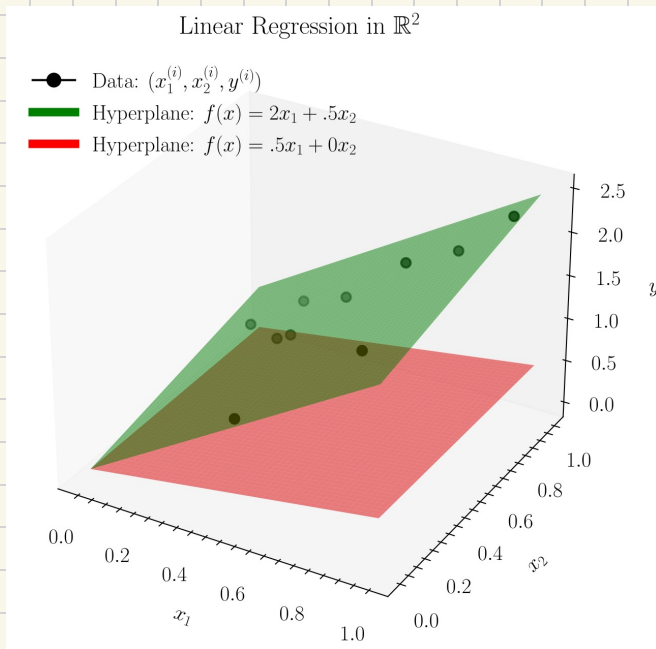
A: Set $\frac{\partial \mathcal{L}(w^*)}{\partial w} = 0$
and solve for w^*

Should get $w^* = \frac{\sum_{i=1}^n y^{(i)} x^{(i)}}{\sum_{i=1}^n (x^{(i)})^2}$

Multivariate Linear Regression

$$x^{(1)}, \dots, x^{(n)} \in \mathbb{R}^d$$

① Model: $f(x) = w^T x = \sum_{j=1}^d w_j x_j$



② MSE Loss!

$$\mathcal{L}(w) = \frac{1}{n} \sum_{i=1}^n (f(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{n} \sum_{i=1}^n (x^{(i)T} w - y^{(i)})^2$$

$$= \frac{1}{n} \|Xw - y\|_2^2$$

$$X = \begin{bmatrix} -x^{(1)T}- \\ \vdots \\ -x^{(n)T}- \end{bmatrix}$$

$n \times d$

$$y = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{bmatrix}$$

$n \times 1$

(3) Exact Optimization

$$\text{Set } \nabla_w \mathcal{L}(w^*) = 0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \text{ and solve for } w^*$$

$$\nabla_w \mathcal{L}(w) = \begin{bmatrix} \frac{\partial \mathcal{L}(w)}{\partial w_1} \\ \vdots \\ \frac{\partial \mathcal{L}(w)}{\partial w_d} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}(w)}{\partial w_i} = \lim_{\Delta \rightarrow 0} \frac{\mathcal{L}(w + \Delta e_i) - \mathcal{L}(w)}{\Delta}$$

$$= \lim_{\Delta \rightarrow 0} \frac{\frac{1}{n} \|X(w + \Delta e_i) - y\|_2^2 - \frac{1}{n} \|Xw - y\|_2^2}{\Delta}$$

$$= \lim_{\Delta \rightarrow 0} \frac{\frac{1}{n} \|Xw - y\|_2^2 - 2(Xe_i)^T(Xw - y) + \Delta^2 \|Xe_i\|_2^2 - \frac{1}{n} \|Xw - y\|_2^2}{\Delta}$$

$$= \lim_{\Delta \rightarrow 0} \frac{1}{n} \left(-2(Xe_i)^T(Xw - y) + \Delta \|Xe_i\|_2^2 \right)$$

$$= -\frac{2}{n} (Xe_i)^T(Xw - y) = -\frac{2}{n} X_{:,i}^T(Xw - y)$$

$$\nabla_w \mathcal{L}(w) = -\frac{2}{n} X^T(Xw - y)$$

$$\begin{aligned} \|a+b\|_2^2 &= (a+b)^T(a+b) \\ &= a^T a + a^T b + b^T a + b^T b \\ &= \|a\|_2^2 + 2a^T b + \|b\|_2^2 \end{aligned}$$

Solve for w^* when

$$\nabla_w \mathcal{L}(w^*) = -\frac{2}{n} X^T (Xw^* - y) = 0$$

$$X^T X w^* - X^T y = 0$$

$$X^T X w^* = X^T y$$

$$w^* = (X^T X)^{-1} X^T y$$

↑ could be
pseudoinverse